In [1]:
```r
library(Seurat)

library(reticulate)
library(anndata)

library(ggplot2)
library(ggpubr)
library(pheatmap)
library(dplyr)
library(tidyr)
library(RColorBrewer)
library(clustree)

library(UpSetR)
library(grid)

library(PRROC)

library(tidyverse)
library(Matrix)

getwd()

dataset_id <- "simulated_mm_RA"
genome_id <- "mm10"
samples <- c("all", "old", "young")

for (sample in samples) {
    dir.create(paste0("figures_", dataset_id, "_", sample))
}

colorTools <- c(
    "STARsolo_TE" = "#FFBC81",
    "STARsolo_TE_EM" = "#f0df93",
    "SoloTE_unique" = "#A4DD9B",
    "SoloTE_thr2" = "#6FC69D",
    "SoloTE_thr1" = "#4BB2BB",
    "SoloTE_thr0" = "#3989BF",
    "Stellarscope" = "#4F5D93",
    "simulated" = "grey70"
)

thrMinCells <- 500 * 0.05
```

```
Loading required package: SeuratObject

Loading required package: sp

‘SeuratObject’ was built under R 4.3.3 but the current version is
4.4.1; it is recomended that you reinstall ‘SeuratObject’ as the ABI
for R may have changed

‘SeuratObject’ was built with package ‘Matrix’ 1.6.4 but the current
version is 1.7.5; it is recomended that you reinstall ‘SeuratObject’ as
the ABI for ‘Matrix’ may have changed


Attaching package: ‘SeuratObject’


The following objects are masked from ‘package:base’:

    intersect, t



Attaching package: ‘anndata’


The following object is masked from ‘package:SeuratObject’:

    Layers


Warning message:
“package ‘dplyr’ was built under R version 4.4.3”

Attaching package: ‘dplyr’


The following objects are masked from ‘package:stats’:
```

```
    filter, lag


The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union


Warning message:
“package ‘tidyr’ was built under R version 4.4.3”
Warning message:
“package ‘RColorBrewer’ was built under R version 4.4.3”
Loading required package: ggraph


Attaching package: ‘ggraph’


The following object is masked from ‘package:sp’:

    geometry


Loading required package: rlang

Warning message:
“package ‘rlang’ was built under R version 4.4.3”
Warning message:
“package ‘tidyverse’ was built under R version 4.4.3”
Warning message:
“package ‘tibble’ was built under R version 4.4.3”
Warning message:
“package ‘readr’ was built under R version 4.4.3”
Warning message:
“package ‘purrr’ was built under R version 4.4.3”
Warning message:
“package ‘stringr’ was built under R version 4.4.3”
Warning message:
“package ‘forcats’ was built under R version 4.4.3”
Warning message:
“package ‘lubridate’ was built under R version 4.4.3”
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ forcats   1.0.1     ✔ readr     2.2.0
✔ lubridate 1.9.5     ✔ stringr   1.6.0
✔ purrr     1.2.2     ✔ tibble    3.3.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter()      masks stats::filter()
✖ purrr::flatten()     masks rlang::flatten()
✖ purrr::flatten_chr() masks rlang::flatten_chr()
✖ purrr::flatten_dbl() masks rlang::flatten_dbl()
✖ purrr::flatten_int() masks rlang::flatten_int()
✖ purrr::flatten_lgl() masks rlang::flatten_lgl()
✖ purrr::flatten_raw() masks rlang::flatten_raw()
✖ purrr::invoke()      masks rlang::invoke()
✖ dplyr::lag()         masks stats::lag()
✖ readr::read_csv()    masks anndata::read_csv()
✖ purrr::splice()      masks rlang::splice()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

Attaching package: ‘Matrix’


The following objects are masked from ‘package:tidyr’:

    expand, pack, unpack

```

'/mnt/TEdata/TEbenchmarking/Rscripts'

```
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):
“'figures_simulated_mm_RA_all' already exists”
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):
“'figures_simulated_mm_RA_old' already exists”
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):
“'figures_simulated_mm_RA_young' already exists”
```

In [ ]:

```
# set paths were results from the snakemake workflow are saved
minnow_path <- "/mnt/TEresults/simulation/minnow"
STARsolo_TE_path <- "/mnt/volume_1p5T"
SoloTE_path <- "/mnt/volume_1p5T"
Stellarscope_path <- "/mnt/TEresults2"
```

In [2]:

```
# import TE annotation with conversion columns for SoloTE and Stellarscope
```

```
conversion_table <- read.table(paste0("annotation/annotation_", genome_id, "_conversion_withAge.tsv"))
conversion_table$soloteID <- gsub("\\?", "", conversion_table$soloteID)
```

Create list of objects with counts generated by each tool

In [3]:
```
objList <- list()

objGeneList <- list()
```

Splatter simulation

In [4]:
```
splatter_objs <- list()
samples <- c("all", "old", "young")

main_path <- paste0(minnow_path, "/build/minnow_simulate_mm_RA_TE_")

for (sample in samples) {
    print(sample)

    path <- paste0(main_path, sample, "/alevin/")

    splatter_mat <- read.table(paste0(path, "quants_mat.csv"),
                               sep = ",", header = FALSE)

    # remove NAs, which are present because rows end with ","
    splatter_mat <- splatter_mat %>% select_if(~ !all(is.na(.)))
    colnames(splatter_mat) <- read.table(paste0(path, "quants_mat_cols.txt"))$V1
    rownames(splatter_mat) <- read.table(paste0(path, "quants_mat_rows.txt"))$V1
    # transpose matrix
    splatter_mat <- t(splatter_mat)

    print("loci in conversion table:")
    print(table(rownames(splatter_mat) %in% conversion_table$locusID))

    rownames(splatter_mat) <- gsub(pattern = "_", replacement = "-", rownames(splatter_mat))

    splatter_objs[[sample]] <- Seurat::CreateSeuratObject(splatter_mat,
        project = paste0("simulated_", sample),
        min.cells = thrMinCells, min.features = 0
    )
    print(splatter_objs[[sample]])
    # around 100 features are removed because expressed in less than the threshold of cells
    print("loci matching stellarscope ID:")
    print(table(rownames(splatter_objs[[sample]]) %in% conversion_table$stellarscopeID))

    print("Summary of n counts")
    print(summary(splatter_objs[[sample]]$nCount_RNA))
    print("Summary of n features")
    print(summary(splatter_objs[[sample]]$nFeature_RNA))
}
```

```
[1] "all"
[1] "loci in conversion table:"

TRUE
5273
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
An object of class Seurat
4947 features across 500 samples within 1 assay
Active assay: RNA (4947 features, 0 variable features)
 1 layer present: counts
[1] "loci matching stellarscope ID:"

TRUE
4947
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  28658   52253   59826   60754   68977  112605
[1] "Summary of n features"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   2819    3408    3528    3521    3649    4036
[1] "old"
[1] "loci in conversion table:"

TRUE
3173
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
```

```
An object of class Seurat 
3030 features across 500 samples within 1 assay 
Active assay: RNA (3030 features, 0 variable features)
 1 layer present: counts
[1] "loci matching stellarscope ID:"

TRUE
3030
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  28191   52205   59970   60769   68261  113835 
[1] "Summary of n features"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   2018    2294    2360    2353    2415    2613 
[1] "young"
[1] "loci in conversion table:"

TRUE
2100
```

```
Warning message:
“Data is of class matrix. Coercing to dgCMatrix.”
```

```
An object of class Seurat 
2030 features across 500 samples within 1 assay 
Active assay: RNA (2030 features, 0 variable features)
 1 layer present: counts
[1] "loci matching stellarscope ID:"

TRUE
2030
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  27752   52153   60062   60785   68740  112572 
[1] "Summary of n features"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   1428    1634    1668    1665    1702    1816 
```

Normalize splatter

In [5]:
```
for (sample in samples) {
    splatter_objs[[sample]] <- NormalizeData(splatter_objs[[sample]],
        normalization.method = "LogNormalize",
        scale.factor = 10000
    )
}
```

```
Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts
```

STARsolo TE

In [6]:
```
STARsolo_TE_objs <- list()
samples <- c("all", "old", "young")

main_path <- paste0(STARsolo_TE_path, "/results/STARsolo_EM_TE/", dataset_id, "/")

for (sample in samples) {
    print(sample)

    path <- paste0(main_path, sample, "/TE_Solo.out/Gene/raw/")

    filteredBarcodes <- read.table(paste0(main_path, sample, "/TE_Solo.out/Gene/filtered/barcodes.tsv"))$V1

    STARsolo_TE_mat <- Seurat::ReadMtx(
        mtx = paste0(path, "matrix.mtx"),
        cells = paste0(path, "barcodes.tsv"),
        features = paste0(path, "features.tsv")
    ) # read matrix
    STARsolo_TE_mat <- STARsolo_TE_mat[, filteredBarcodes] # keep only barcodes passing thresholds

    STARsolo_TE_objs[[sample]] <- Seurat::CreateSeuratObject(STARsolo_TE_mat,
        project = paste0("STARsolo_TE_", sample),
        min.cells = thrMinCells, min.features = 0
    )

    # Remove some classes of TEs (already not present in SoloTE)
```

```
    classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
    starsoloTEs <- Features(STARsolo_TE_objs[[sample]])

    filteredStarsoloTEs <- conversion_table[conversion_table$stellarscopeID %in% starsoloTEs, ] %>%
        dplyr::filter(!class %in% classesToExclude) %>%
        pull(stellarscopeID)

    print("Percentage of TEs in removed orders:")
    print(length(setdiff(starsoloTEs, filteredStarsoloTEs)) / length(starsoloTEs) * 100)

    STARsolo_TE_objs[[sample]] <- STARsolo_TE_objs[[sample]][intersect(starsoloTEs, filteredStarsoloTEs), ]

    print("Summary of n counts")
    summary(STARsolo_TE_objs[[sample]]$nCount_RNA)
    print("Summary of n features")
    summary(STARsolo_TE_objs[[sample]]$nFeature_RNA)

    print("Loci present in the annotation:")
    print(table(rownames(STARsolo_TE_objs[[sample]]) %in% conversion_table$stellarscopeID))
}

# add to the list of objects
objList[["STARsolo_TE"]] <- STARsolo_TE_objs
```

```
[1] "all"
Warning message:
“Feature names cannot have underscores ('_'), replacing with dashes ('-')”
[1] "Percentage of TEs in removed orders:"
[1] 0
[1] "Summary of n counts"
[1] "Summary of n features"
[1] "Loci present in the annotation:"

 TRUE
14107
[1] "old"
Warning message:
“Feature names cannot have underscores ('_'), replacing with dashes ('-')”
[1] "Percentage of TEs in removed orders:"
[1] 0
[1] "Summary of n counts"
[1] "Summary of n features"
[1] "Loci present in the annotation:"

TRUE
3006
[1] "young"
Warning message:
“Feature names cannot have underscores ('_'), replacing with dashes ('-')”
[1] "Percentage of TEs in removed orders:"
[1] 0
[1] "Summary of n counts"
[1] "Summary of n features"
[1] "Loci present in the annotation:"

 TRUE
17410
```

STARsolo TE with EM

In [7]:

```
STARsolo_TE_EM_objs <- list()
samples <- c("all", "old", "young")

main_path <- paste0("/results/STARsolo_EM_TE/", dataset_id, "/")

for (sample in samples) {
    print(sample)

    path <- paste0(main_path, sample, "/TE_Solo.out/Gene/raw/")
    filteredBarcodes <- read.table(paste0(main_path, sample, "/TE_Solo.out/Gene/filtered/barcodes.tsv"))$V1

    STARsolo_TE_EM_mat <- Seurat::ReadMtx(
        mtx = paste0(path, "UniqueAndMult-EM.mtx"),
        cells = paste0(path, "barcodes.tsv"),
        features = paste0(path, "features.tsv")
    ) # read matrix
    STARsolo_TE_EM_mat <- STARsolo_TE_EM_mat[, filteredBarcodes] # keep only barcodes passing thresholds

    STARsolo_TE_EM_objs[[sample]] <- Seurat::CreateSeuratObject(STARsolo_TE_EM_mat,
        project = paste0("STARsolo_TE_EM_", sample),
        min.cells = thrMinCells, min.features = 0
```

```
    )

    # Remove some classes of TEs (already not present in SoloTE)
    classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
    starsoloTEs <- Features(STARsolo_TE_EM_objs[[sample]])

    filteredStarsoloTEs <- conversion_table[conversion_table$stellarscopeID %in% starsoloTEs, ] %>%
        dplyr::filter(!class %in% classesToExclude) %>%
        pull(stellarscopeID)

    print("Percentage of TEs in removed orders:")
    print(length(setdiff(starsoloTEs, filteredStarsoloTEs)) / length(starsoloTEs) * 100)

    STARsolo_TE_EM_objs[[sample]] <- STARsolo_TE_EM_objs[[sample]][intersect(starsoloTEs, filteredStarsoloTEs),

    print("Summary of n counts")
    print(summary(STARsolo_TE_EM_objs[[sample]]$nCount_RNA))
    print("Summary of n feature")
    print(summary(STARsolo_TE_EM_objs[[sample]]$nFeature_RNA))

    print("Loci present in the annotation:")
    print(table(rownames(STARsolo_TE_EM_objs[[sample]]) %in% conversion_table$stellarscopeID))
}

# add to the list of objects
objList[["STARsolo_TE_EM"]] <- STARsolo_TE_EM_objs
```

```
[1] "all"
Warning message:
“Feature names cannot have underscores ('_'), replacing with dashes ('-')”
[1] "Percentage of TEs in removed orders:"
[1] 0.005746301
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  29696   49460   55574   56187   62737   97257
[1] "Summary of n feature"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   6268    7458    7701    7704    7976    9143
[1] "Loci present in the annotation:"

 TRUE
34803
[1] "old"
Warning message:
“Feature names cannot have underscores ('_'), replacing with dashes ('-')”
[1] "Percentage of TEs in removed orders:"
[1] 0.03131851
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  28444   52263   59914   60686   68063  112737
[1] "Summary of n feature"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   2088    2362    2434    2427    2493    2687
[1] "Loci present in the annotation:"

TRUE
3192
[1] "young"
Warning message:
“Feature names cannot have underscores ('_'), replacing with dashes ('-')”
[1] "Percentage of TEs in removed orders:"
[1] 0
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  30052   45604   49923   50030   54333   77213
[1] "Summary of n feature"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   7392    8256    8474    8463    8697    9818
[1] "Loci present in the annotation:"

 TRUE
37855
```

SoloTE Default (thr 255)

In [8]:

```
SoloTE_TEobjs <- list()
samples <- c("all", "old", "young")

main_path <- paste0(SoloTE_path, "/results/SoloTEout/", dataset_id, "/")

for (sample in samples) {
```

```r
    print(sample)
    path <- paste0(main_path, sample, "/", sample, "_SoloTE_output/", sample, "_legacytes_MATRIX/")
    # read matrix
    SoloTE_mat <- Seurat::ReadMtx(
        mtx = paste0(path, "matrix.mtx"),
        cells = paste0(path, "barcodes.tsv"),
        features = paste0(path, "features.tsv")
    )
    # filter cells
    filteredBarcodes <- colnames(splatter_objs[[sample]])
    filteredBarcodes %in% colnames(SoloTE_mat)
    SoloTE_mat <- SoloTE_mat[,filteredBarcodes]
    # select locus level TEs
    TEs <- grep("SoloTE", rownames(SoloTE_mat), value = T)
    locusTEs <- grep("chr", TEs, value = T)
    # select genes
    genes <- setdiff(rownames(SoloTE_mat), TEs)

    # subset the matrix keeping only genes
    SoloTE_GENEmat <- SoloTE_mat[genes, ]
    # create Seurat object with shallow filtering of genes expressed in at least 50 cells and cells expressing a
    SoloTE_GENEobj <- Seurat::CreateSeuratObject(SoloTE_GENEmat,
        project = paste0("SoloTE_unique_", sample),
        min.cells = thrMinCells, min.features = 0
    )
    print("gene obj")
    print(SoloTE_GENEobj)
    #209 genes

    # subset the matrix keeping only TEs
    SoloTE_TEmat <- SoloTE_mat[locusTEs, ]
    # remove "SoloTE" from the name of the TEs and subtitute "|" and "_" with "-"
    rownames(SoloTE_TEmat) <- gsub("SoloTE\\|", "", rownames(SoloTE_TEmat))
    rownames(SoloTE_TEmat) <- gsub("\\|", "-", rownames(SoloTE_TEmat))
    rownames(SoloTE_TEmat) <- gsub("\\_", "-", rownames(SoloTE_TEmat))
    rownames(SoloTE_TEmat) <- gsub("\\?", "", rownames(SoloTE_TEmat))

    table(rownames(SoloTE_TEmat) %in% conversion_table$soloteID)
    # transform into Stellarscope IDs
    rownames(SoloTE_TEmat) <- conversion_table$stellarscopeID[match(rownames(SoloTE_TEmat), conversion_table$so

    # create Seurat object with shallow filtering of TEs expressed in at least thrMinCells cells and cells expre
    SoloTE_TEobjs[[sample]] <- Seurat::CreateSeuratObject(SoloTE_TEmat,
        project = paste0("SoloTE_unique_", sample),
        min.cells = thrMinCells, min.features = 0
    )

    print("Summary of n counts")
    print(summary(SoloTE_TEobjs[[sample]]$nCount_RNA))
    print("Summary of n features")
    print(summary(SoloTE_TEobjs[[sample]]$nFeature_RNA))

    print("Loci present in the annotation:")
    print(table(rownames(SoloTE_TEobjs[[sample]]) %in% conversion_table$stellarscopeID))
}

objList[["SoloTE_unique"]] <- SoloTE_TEobjs
```

```
[1] "all"
[1] "gene obj"
An object of class Seurat 
154 features across 500 samples within 1 assay 
Active assay: RNA (154 features, 0 variable features)
 1 layer present: counts
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  20778   34062   38288   38732   42958   67021 
[1] "Summary of n features"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   2281    2634    2712    2708    2788    3062 
[1] "Loci present in the annotation:"

TRUE 
6128 
[1] "old"
[1] "gene obj"
An object of class Seurat 
22 features across 500 samples within 1 assay 
Active assay: RNA (22 features, 0 variable features)
 1 layer present: counts
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  32854   55496   62526   63211   70242  112164 
[1] "Summary of n features"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   2017    2279    2345    2337    2399    2591 
[1] "Loci present in the annotation:"

TRUE 
3034 
[1] "young"
[1] "gene obj"
An object of class Seurat 
208 features across 500 samples within 1 assay 
Active assay: RNA (208 features, 0 variable features)
 1 layer present: counts
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   5399    7042    7602    7610    8118   11176 
[1] "Summary of n features"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   1019    1072    1090    1091    1110    1186 
[1] "Loci present in the annotation:"

TRUE 
4935 
```

SoloTE thr 2,1 and 0

In [9]:

```
SoloTE_TEobjs <- list()
samples <- c("all", "old", "young")

for(thr in c("thr2","thr1","thr0")){

    main_path <- paste0(SoloTE_path, "/results/SoloTEout_",thr,"/", dataset_id, "/")

    for (sample in samples){
        print(sample)
        path <- paste0(main_path, sample, "/", sample, "_SoloTE_output/", sample, "_legacytes_MATRIX/")
        # read matrix
        SoloTE_mat <- Seurat::ReadMtx(
            mtx = paste0(path, "matrix.mtx"),
            cells = paste0(path, "barcodes.tsv"),
            features = paste0(path, "features.tsv")
        )
        # filter cells
        filteredBarcodes <- colnames(splatter_objs[[sample]])

        SoloTE_mat <- SoloTE_mat[,filteredBarcodes]
        # select locus level TEs
        TEs <- grep("SoloTE", rownames(SoloTE_mat), value = T)
        locusTEs <- grep("chr", TEs, value = T)
        # select genes
        genes <- setdiff(rownames(SoloTE_mat), TEs)

        # subset the matrix keeping only genes
        SoloTE_GENEmat <- SoloTE_mat[genes, ]
        # create Seurat object with shallow filtering of genes expressed in at least 50 cells and cells express
        SoloTE_GENEobj <- Seurat::CreateSeuratObject(SoloTE_GENEmat,
```

```
            project = paste0("SoloTE_",thr,"_", sample),
            min.cells = thrMinCells, min.features = 0
        )
        #209 genes

        # subset the matrix keeping only TEs
        SoloTE_TEmat <- SoloTE_mat[locusTEs, ]
        # remove "SoloTE" from the name of the TEs and subtitute "|" and "_" with "-"
        rownames(SoloTE_TEmat) <- gsub("SoloTE\\|", "", rownames(SoloTE_TEmat))
        rownames(SoloTE_TEmat) <- gsub("\\|", "-", rownames(SoloTE_TEmat))
        rownames(SoloTE_TEmat) <- gsub("\\_", "-", rownames(SoloTE_TEmat))
        rownames(SoloTE_TEmat) <- gsub("\\?", "", rownames(SoloTE_TEmat))

        table(rownames(SoloTE_TEmat) %in% conversion_table$soloteID)
        # transform into Stellarscope IDs
        rownames(SoloTE_TEmat) <- conversion_table$stellarscopeID[match(rownames(SoloTE_TEmat), conversion_tabl

        # create Seurat object with shallow filtering of TEs expressed in at least thrMinCells cells and cells 
        SoloTE_TEobjs[[sample]] <- Seurat::CreateSeuratObject(SoloTE_TEmat,
            project = paste0("SoloTE_",thr,"_", sample),
            min.cells = thrMinCells, min.features = 0
        )

        summary(SoloTE_TEobjs[[sample]]$nCount_RNA)
        summary(SoloTE_TEobjs[[sample]]$nFeature_RNA)

        print("Loci present in the annotation:")
        print(table(rownames(SoloTE_TEobjs[[sample]]) %in% conversion_table$stellarscopeID))
    }

    objList[[paste0("SoloTE_",thr)]] <- SoloTE_TEobjs
}
```

```
[1] "all"
[1] "Loci present in the annotation:"

TRUE
8675
[1] "old"
[1] "Loci present in the annotation:"

TRUE
3059
[1] "young"
[1] "Loci present in the annotation:"

TRUE
8637
[1] "all"
[1] "Loci present in the annotation:"

 TRUE
11686
[1] "old"
[1] "Loci present in the annotation:"

TRUE
3078
[1] "young"
[1] "Loci present in the annotation:"

 TRUE
13116
[1] "all"
[1] "Loci present in the annotation:"

 TRUE
33662
[1] "old"
[1] "Loci present in the annotation:"

TRUE
3271
[1] "young"
[1] "Loci present in the annotation:"

 TRUE
42536
```

Stellarscope

In [10]:
```
Stellarscope_objs <- list()
```

```
samples <- c("all", "old", "young")

main_path <- paste0(stellarscope_results, "/snakemake_results/results/stellarscope_out/", dataset_id, "/")

for (sample in samples){
    print(sample)
    path <- paste0(main_path, sample, "/pseudobulk/")
    # read matrix
    Stellarscope_mat <- Seurat::ReadMtx(mtx = paste0(path, sample, "_pseudobulk-TE_counts.mtx"),
                                cells = paste0(path, sample,"_pseudobulk-barcodes.tsv"),
                                features = paste0(path, sample,"_pseudobulk-features.tsv"),
                                feature.column = 1) # read matrix
    # remove the unassigned counts
    no_feature <- Stellarscope_mat["__no_feature",]
    Stellarscope_mat <- Stellarscope_mat[setdiff(rownames(Stellarscope_mat),"__no_feature"),]
    # substitute "_" with "-"
    rownames(Stellarscope_mat) <- gsub(pattern = "_", replacement = "-", rownames(Stellarscope_mat))

    # filter cells passing filters
    filteredBarcodes <- colnames(splatter_objs[[sample]])
    Stellarscope_mat <- Stellarscope_mat[,filteredBarcodes]

    # create Seurat object with shallow filtering of TEs expressed in at least 5% of cells and cells expressing
    Stellarscope_objs[[sample]] <- Seurat::CreateSeuratObject(Stellarscope_mat, project = paste0("Stellarscope_
                            min.cells = thrMinCells, min.features = 0)

    # Remove some classes of TEs (already not present in SoloTE)
    classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
    stellarscopeTEs <- Features(Stellarscope_objs[[sample]])

    filteredStellarscopeTEs <- conversion_table[conversion_table$stellarscopeID %in% stellarscopeTEs,] %>%
        filter(! class %in% classesToExclude) %>% pull(stellarscopeID)
    print("percentage of loci removed")
    print(length(setdiff(stellarscopeTEs, filteredStellarscopeTEs))/ length(stellarscopeTEs) * 100) # percentage

    Stellarscope_objs[[sample]]  <- Stellarscope_objs[[sample]][intersect(stellarscopeTEs, filteredStellarscope

    print("n count summary")
    print(summary(Stellarscope_objs[[sample]]$nCount_RNA))
    print("n features summary")
    print(summary(Stellarscope_objs[[sample]]$nFeature_RNA))

    print("Loci present in the annotation:")
    print(table(rownames(Stellarscope_objs[[sample]]) %in% conversion_table$stellarscopeID))
}

objList[["Stellarscope"]] <- Stellarscope_objs
```

```
[1] "all"
[1] "percentage of loci removed"
[1] 0.01145147
[1] "n count summary"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  30750   47662   52945   53471   58916   89185 
[1] "n features summary"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   5351    5834    5936    5930    6032    6627 
[1] "Loci present in the annotation:"

 TRUE 
17463 
[1] "old"
[1] "percentage of loci removed"
[1] 0.03269042
[1] "n count summary"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  32852   55539   62736   63338   70400  112476 
[1] "n features summary"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   2010    2284    2348    2342    2404    2598 
[1] "Loci present in the annotation:"

TRUE 
3058 
[1] "young"
[1] "percentage of loci removed"
[1] 0
[1] "n count summary"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  29311   40844   44180   44270   47521   66637 
[1] "n features summary"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   5926    6098    6140    6136    6176    6569 
[1] "Loci present in the annotation:"

 TRUE 
18728 
```

Normalize objects

In [11]:

```
for (sample in samples){
    for (tool in names(objList)){
        objList[[tool]][[sample]] <- NormalizeData(objList[[tool]][[sample]],
            normalization.method = "LogNormalize", scale.factor = 10000)
    }
}
```

```
Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts
```

```
Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

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Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts
```

Save objects

In [12]:

```
dir.create("workspaces")
save.image("workspaces/00_evaluation_objectCreation.Rdata")
```

```
Warning message in dir.create("workspaces"):
“'workspaces' already exists”
```